

MAC 2312 — CALCULUS II

Lecture 6: Trigonometric Integrals

Wednesday, September 16 • Sections 3.2

• Guided Notes •

How to Use These Guided Notes: These guided notes are designed to help you organize the main ideas from each lecture but

1. we will not complete every example in class.
2. Although most definitions and theorems are included, you may need to add additional notes, examples, or explanations during class.
3. Occasionally, you will see sections labeled **Deeper Insight**. These sections provide interesting background information and help develop a deeper understanding of calculus, but they are not part of the core Calculus II content.

When you first encounter trigonometric integrals, the hardest part is often deciding *which method to use*. The purpose of this guide is to help you quickly recognize the structure of an integral and choose an efficient strategy. The techniques in this handout are not new integration methods. They are simply ways of using identities, substitution, and integration by parts to rewrite a problem into a form we already know how to integrate.

Theorem 6.1 – Important Trigonometric Properties

Definitions

$$\tan(x) = \frac{\sin(x)}{\cos(x)}, \quad \sec(x) = \frac{1}{\cos(x)}, \quad \csc(x) = \frac{1}{\sin(x)}.$$

Most important identities

$$\sin^2(x) + \cos^2(x) = 1, \quad \tan^2(x) + 1 = \sec^2(x).$$

Double-angle identities:

$$\sin(2x) = 2 \sin(x) \cos(x), \quad \cos(2x) = \cos^2(x) - \sin^2(x).$$

Half-angle identities:

$$\sin^2(x) = \frac{1 - \cos(2x)}{2}, \quad \cos^2(x) = \frac{1 + \cos(2x)}{2}.$$

Less common antiderivatives

$$\int \tan(x) \, dx = -\ln |\cos(x)| + C, \quad \int \sec(x) \, dx = \ln |\sec(x) + \tan(x)| + C.$$

$$\int \sec^2(x) \, dx = \tan(x) + C, \quad \int \sec(x) \tan(x) \, dx = \sec(x) + C.$$

Products and powers of sine and cosine

Suppose you need to evaluate $\int \sin^m(x) \cos^n(x) dx$, where m and n are nonnegative integers. Thus, m is the exponent of sine and n is the exponent of cosine.

Situation	Recommended Strategy
Power of sine is odd	Save one factor of $\sin(x)$ and use $u = \cos(x)$.
Power of cosine is odd	Save one factor of $\cos(x)$ and use $u = \sin(x)$.
Both powers are even	Apply the half-angle identities.

More detailed steps can be found in table 1.

Quick decision guide

Ask the following questions:

1. Is the power of sine odd?
 - Yes $\Rightarrow$ save one factor of $\sin(x)$ and rewrite the remaining sine powers using $\sin^2(x) = 1 - \cos^2(x)$. Then apply the substitution rule with $u = \cos(x)$.
 - No $\Rightarrow$ continue.
2. Is the power of cosine odd?
 - Yes $\Rightarrow$ save one factor of $\cos(x)$ and rewrite the remaining cosine powers using $\cos^2(x) = 1 - \sin^2(x)$. Then apply the substitution $u = \sin(x)$.
 - No $\Rightarrow$ continue.
3. Both powers are even.
 - Use the half-angle identities.

Understanding the Goal

Before using the decision guide, let us examine why each strategy works.

Odd power of sine: save one factor of sine

Goal: Create a factor of $\sin(x) dx$ so that a substitution with $u = \cos(x)$ can be used after rewriting the remaining even power of sine in terms of cosine using the identity $\sin^2(x) = 1 - \cos^2(x)$.

Example 6.1

$$\int \sin^3(x) \cos^2(x) \, dx = \int \sin^2(x) \cos^2(x) \sin(x) \, dx \quad (\text{saving one factor of } \sin(x))$$

$$=$$
Odd power of cosine: save one factor of cosine

Goal: Similar to the previous case, but instead of creating a factor of $\sin(x) \, dx$, we create a factor of $\cos(x) \, dx$ so that $u = \sin(x)$ can be used after rewriting the remaining even power of cosine in terms of sine using the identity $\cos^2(x) = 1 - \sin^2(x)$.

Both powers are even: use half-angle identities

Goal: Use half-angle identities to reduce the exponents.

Example 6.2

Let us determine

$$\int \sin^2(x) \cos^2(x) \, dx.$$

Using the half-angle identities, it can be rewritten as

$$\int \sin^2(x) \cos^2(x) \, dx = \int \left(\frac{1 - \cos(2x)}{2} \right) \left(\frac{1 + \cos(2x)}{2} \right) \, dx.$$

After expansion, the resulting integral is easier to evaluate.

$$\int \frac{1 - \cos 2x}{2} \cdot \frac{1 + \cos 2x}{2} dx = \int \frac{1 - \cos^2 2x}{4} dx \quad (\text{expanding})$$

$$=$$

A second alternative to solve this integral is using the identity $\sin(2x) = 2 \sin(x) \cos(x)$ before applying the half-angle identities.

Note that

$$\sin(2x) = 2 \sin(x) \cos(x) \Rightarrow \sin^2(2x) = 4 \sin^2(x) \cos^2(x) \Rightarrow \frac{1}{4} \sin^2(2x) = \sin^2(x) \cos^2(x).$$

Then,

$$\begin{aligned} \int \sin^2 x \cos^2 x dx &= \frac{1}{4} \int \sin^2(2x) dx \quad (\text{using identity above}) \\ &= \frac{1}{4} \int \frac{1}{2} (1 - \cos(4x)) dx \quad (\text{using half-angle identities}) \\ &= \end{aligned}$$

How Do We Check Our Answers?

One of the best habits in Calculus II is to verify your answer by differentiating. If

$$\int f(x) dx = F(x) + C,$$

then your answer is correct if $F'(x) = f(x)$. After finding an antiderivative: differentiate your final answer, simplify, and verify that the result matches the original integrand. If the derivative matches the original integrand, your answer is correct.

Products and powers of secant and tangent

Suppose you need to evaluate $\int \tan^m(x) \sec^n(x) dx$, where m and n are nonnegative integers. Thus, m is the exponent of tangent and n is the exponent of secant.

Situation	Recommended Strategy
Even power of secant	Save a factor of $\sec^2(x)$.
Odd power of tangent, at least one secant	Save a factor of $\sec(x) \tan(x)$.
Odd power of secant, no tangent	Use integration by parts to reduce the secant exponent.
Just tangent	Convert using $\tan^2(x) = \sec^2(x) - 1$.
Even power of tangent, odd power of secant	Rewrite using $\tan^2(x) = \sec^2(x) - 1$ to obtain a sum of integrals with odd powers of secant only.

More detailed steps can be found in tables 2 and 3.

The techniques used for tangent and secant are designed to create one of these derivative patterns.

- An extra factor of $\sec^2(x)$ suggests $u = \tan(x)$.
- An extra factor of $\sec(x) \tan(x)$ suggests $u = \sec(x)$.
- Even powers of tangent can often be rewritten using $\tan^2(x) = \sec^2(x) - 1$.

Quick Decision Guide

Ask the following questions:

1. Is the exponent of secant even and at least 2?
 - Yes $\Rightarrow$ save a factor of $\sec^2(x)$, rewrite the remaining secant powers using $\sec^2(x) = \tan^2(x) + 1$, and use $u = \tan(x)$.
 - No $\Rightarrow$ continue.
2. Is the exponent of tangent odd, with at least one factor of secant?
 - Yes $\Rightarrow$ save one factor of $\sec(x) \tan(x)$ and rewrite the remaining powers of tangent using $\tan^2(x) = \sec^2(x) - 1$. Then apply the substitution rule with $u = \sec(x)$.
 - No $\Rightarrow$ continue.
3. Is the secant exponent odd, with no tangent factor (integrals of the form $\int \sec^n(x) dx$ with n odd)?
 - Yes $\Rightarrow$ if secant exponent is 1, we already know one antiderivative. Otherwise, save one factor of $\sec^2(x)$, apply integration by parts with $dv = \sec^2(x) dx$, and use $\tan^2(x) = \sec^2(x) - 1$ in the resulting integral. Combine like-terms.

- No $\Rightarrow$ continue.
4. Does the integral contain only a power of tangent (integrals of the form $\int \tan^m(x) dx$)?
- Yes $\Rightarrow$ if the tangent exponent is 1, we know one antiderivative. If the tangent exponent is larger than 1, save one factor of $\tan^2(x)$, rewrite it as $\sec^2(x) - 1$ and evaluate the resulting integrals.
 - No $\Rightarrow$ continue.
5. Is the power of tangent even and the power of secant odd?
- Yes $\Rightarrow$ use $\tan^2(x) = \sec^2(x) - 1$ to rewrite the integral as a sum of integrals with odd powers of secant and no tangent.
 - No $\Rightarrow$ continue.

Understanding the Goal

Even power of secant: save one factor of $\sec^2(x)$

Goal: Create a factor of $\sec^2(x) dx$ so that $u = \tan(x)$ can be used after transforming the remaining even powers of secant using the identity $\sec^2(x) = \tan^2(x) + 1$.

Example 6.3

$$\int \sec^4(x) dx = \int \sec^2(x) \sec^2(x) dx \text{ (saving one factor of } \sec^2(x))$$

$$=$$

Odd power of tangent and at least one secant

Goal: Create a factor of $\sec(x) \tan(x) dx$ so that $u = \sec(x)$ can be used after transforming the remaining even powers of tangent using the identity $\tan^2(x) = \sec^2(x) - 1$.

Example 6.4

$$\begin{aligned} & \int \tan^3(x) \sec^3(x) \, dx \\ &= \int \tan^2(x) \sec^2(x) \tan(x) \sec(x) \, dx \text{ (saving one factor of } \sec(x) \tan(x)) \\ &= \end{aligned}$$

Odd power of secant, no tangent: use integration by parts

Goal: Save one factor of $\sec^2(x)$ and use integration by parts to reduce the secant exponent.

Example 6.5

$$\int \sec^3(x) \, dx =$$

No secant: use $\tan^2(x) = \sec^2(x) - 1$

Goal: Use $\tan^2(x) = \sec^2(x) - 1$ to transform the integral into one with an even secant exponent and reduce exponents.

Example 6.6

$$\int \tan^4(x) \, dx = \int \tan^2(x) \tan^2(x) \, dx \text{ (saving one factor of } \tan^2(x) \text{)}$$

Let us check the previous answer.

The same steps can be applied if the tangent exponent is odd.

Example 6.7

$$\int \tan^3(x) \, dx = \int \tan(x) \tan^2(x) \, dx$$

Even power of tangent, odd power of secant: use $\tan^2(x) = \sec^2(x) - 1$

Goal: Use $\tan^2(x) = \sec^2(x) - 1$ to transform the integral into a sum of integrals with odd powers of secant.

Example 6.8

$$\begin{aligned}\int \tan^4(x) \sec^3(x) \, dx &= \int (\tan^2(x))^2 \sec^3(x) \, dx \text{ (rewriting the even power of tangent)} \\ &= \int (\sec^2(x) - 1)^2 \sec^3(x) \, dx \text{ (using } \tan^2(x) = \sec^2(x) - 1) \\ &= \end{aligned}$$

The resulting integral only has odd powers of secant.

In an example above, we wrote

$$\int \sec^3(x) \, dx = \frac{1}{2} \sec(x) \tan(x) + \frac{1}{2} \ln |\sec(x) + \tan(x)| + C.$$

To integrate other odd powers of secant, like $\sec^5(x)$ or $\sec^7(x)$, we follow the same steps as before (saving one factor of $\sec^2(x)$, applying integration by parts, combining like-terms).

More detailed tables for each strategy

Case	Technique
m is odd; n any	1. Save one factor of $\sin x$. 2. Rewrite the remaining even power of sine using $\sin^2 x = 1 - \cos^2 x$. 3. Substitute $u = \cos x$, $du = -\sin x \, dx$.
n is odd; m any	1. Save one factor of $\cos x$. 2. Rewrite the remaining even power of cosine using $\cos^2 x = 1 - \sin^2 x$. 3. Substitute $u = \sin x$, $du = \cos x \, dx$.
m and n are both even	1. No factor can be saved for a u-substitution. 2. Use the power-reducing identities $\sin^2 x = \frac{1 - \cos 2x}{2}, \quad \cos^2 x = \frac{1 + \cos 2x}{2}$ to rewrite the integrand. 3. Expand and integrate term by term (repeat as needed).

Table 1: Calculating $\int \sin^m(x) \cos^n(x) \, dx$

Case	Technique
n is even, $n \geq 2$; m any	1. Save a factor of $\sec^2 x$. 2. Rewrite the remaining even power of secant using $\sec^2 x = 1 + \tan^2 x$. 3. Substitute $u = \tan x$, $du = \sec^2 x \, dx$.
m is odd, $m \geq 1$; $n \geq 1$ any	1. Save a factor of $\sec x \tan x$. 2. Rewrite the remaining even power of tangent using $\tan^2 x = \sec^2 x - 1$. 3. Substitute $u = \sec x$, $du = \sec x \tan x \, dx$.

Table 2: Calculating $\int \tan^m(x) \sec^n(x) \, dx$, more common cases

Case	Technique
n is odd, $n \geq 1$; $m = 0$	Suppose the integral we have to solve is $\int \sec^{2j+1} dx$ with $j \geq 0$ (because n is odd). If $j = 0$, we already know one antiderivative. If $j \geq 1$, follow these steps 1. Write $\sec^{2j+1}(x)$ as $\sec^{2j-1} \sec^2(x)$. 2. Apply integration by parts with $u = \sec^{2j-1}(x)$, $dv = \sec^2(x) dx$. 3. Use $\tan^2(x) = \sec^2(x) - 1$. 4. Combine like-terms.
$n = 0$; m any	We already know $\int \tan(x) dx = -\ln \cos(x) + C$. If $m \geq 2$: 1. Save one factor of $\tan^2(x)$. 2. Replace it with $\sec^2(x) - 1$. 3. The new two integrals are an integral with an even power of secant and an integral like this one with a smaller exponent for tangent or a known antiderivative.
m is even and n is odd	1. Rewrite each factor of $\tan^2(x)$ as $\sec^2(x) - 1$. 2. The new integrals are integrals of odd powers of secant (first case in this table).

Table 3: Calculating $\int \tan^m(x) \sec^n(x) dx$, less usual cases

Let us solve some practice problems while avoiding the following common mistakes.

! Common Mistakes

- Choosing the wrong factor to save when integrating products of powers of $\sin(x)$ and $\cos(x)$.
- Incorrectly applying $\sin^2(x) + \cos^2(x) = 1$.
- Forgetting the negative sign when doing $u = \cos(x)$.
- Writing incorrect half-angle identities.
- Choosing the wrong factor to save when integrating products of powers of $\tan(x)$ and $\sec(x)$.
- Forgetting to account for the derivative of the inner angle when the trigonometric function has an argument such as $2x$ or $5t$.
- Incorrectly applying $\tan^2(x) + 1 = \sec^2(x)$.

- Forgetting that integrals involving only powers of $\sec(x)$ may require integration by parts.
 - Forgetting the constant of integration $+C$ when solving indefinite integrals.
-

Practice Problems

Level I problems

1. Determine

- | | |
|---------------------------------------|---|
| (a) $\int \cos^3(x) \, dx.$ | (f) $\int \sin^5(2t) \cos^2(2t) \, dt.$ |
| (b) $\int \sin^5(x) \cos^2(x) \, dx.$ | (g) $\int \cos^3\left(\frac{t}{2}\right) \sin^2\left(\frac{t}{2}\right) \, dt.$ |
| (c) $\int_0^\pi \sin^2(x) \, dx.$ | (h) $\int \sin^2(x) \sin(2x) \, dx.$ |
| (d) $\int \sin^3(x) \cos^2(x) \, dx.$ | (i) $\int \sin(x) \cos\left(\frac{x}{2}\right) \, dx.$ |
| (e) $\int_0^\pi \cos^4(2t) \, dt.$ | (j) $\int_0^{\frac{\pi}{2}} (2 - \sin(\theta))^2 \, d\theta.$ |

2. Determine

- | | |
|--|--|
| (a) $\int \tan(x) \sec^3(x) \, dx.$ | (h) $\int_0^{\frac{\pi}{4}} \tan^3(t) \, dt.$ |
| (b) $\int \tan^2(x) \, dx.$ | (i) $\int \tan^6(x) \sec^4(x) \, dx.$ |
| (c) $\int \tan^4(x) \sec^6(x) \, dx.$ | (j) $\int \tan^4(x) \sec^4(x) \, dx.$ |
| (d) $\int \tan^3(x) \sec(x) \, dx.$ | (k) $\int \tan^2(x) \sec^4(x) \, dx.$ |
| (e) $\int \tan^3(x) \sec^6(x) \, dx.$ | (l) $\int \tan^5(x) \sec^3(x) \, dx.$ |
| (f) $\int_0^{\frac{\pi}{4}} \sec^6(\theta) \tan^6(\theta) \, d\theta.$ | (m) $\int \sec^4\left(\frac{x}{2}\right) \, dx.$ |
| (g) $\int_0^{\frac{\pi}{4}} \tan^4(t) \, dt.$ | (n) $\int \frac{1}{\sec(x)} \, dx.$ |

3. After a substitution, the integral $\int \tan^3(x) \sec^3(x) \, dx$ is equivalent to

A. $\int u^2(u^2 + 1)du.$

C. $\int u^6 du.$

B. $\int u^4 du.$

D. $\int u^2(u^2 - 1)du.$

4. After a substitution, the integral $\int \tan^2(x) \sec^4(x) dx$ is equivalent to

A. $\int u^2(u^2 + 1)du.$

C. $\int u^6 du.$

B. $\int u^4 du.$

D. $\int u^2(u^2 - 1)du.$

5. After a substitution, the integral $\int \tan(x) \sec^5(x) dx$ is equivalent to

A. $\int u^2(u^2 + 1)du.$

C. $\int u^6 du.$

B. $\int u^4 du.$

D. $\int u^2(u^2 - 1)du.$

Practice Problems

Level II problems

1. Find the area enclosed by the graphs of $f(x) = \cos^2(x)$ and $g(x) = \cos(x)$ over the interval $[-\pi, \pi]$.

2. Determine

(a) $\int \sin(x) \sec^5(x) dx.$

(c) $\int t \sin^2(t) dt.$

(b) $\int \sqrt{\cos(\theta)} \sin^3(\theta) d\theta.$

(d) $\int_0^{\frac{\pi}{2}} \cos(5t) \cos(10t) dt.$

3. Find the volume obtained by rotating the region bounded by the curves about the given axis.

(a) $y = \sin(x), y = 0, \frac{\pi}{2} \leq x \leq \pi$; about the x -axis.

(c) $y = \sin(x), y = \cos(x), 0 \leq x \leq \frac{\pi}{4}$; about $y = 1$.

(b) $y = \sin^2(x), y = 0, 0 \leq x \leq \pi$; about the x -axis.

(d) $y = \sec(x), y = \cos(x), 0 \leq x \leq \frac{\pi}{3}$; about $y = -1$.

Remark: You may use a graphing calculator to understand the rotated regions and set up the integrals.

By the end of today you will be able to:

- Classify products involving powers of sine, cosine, tangent, and secant.
- Choose a strategy based on the exponents and available derivative patterns.
- Rewrite a trigonometric integral using an appropriate identity.
- Create a suitable u -substitution by saving the correct factor.
- Recognize when integration by parts is needed for an odd power of secant.
- Verify an antiderivative by differentiating and simplifying the result.